

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : SOLITAIRE

Other means of identification : Not applicable.

Recommended use : Manual Warewashing Detergent

Product dilution information : 0.03 % - 0.08 %

Company

Ecolab Gulf FZE
P.O. Box 17063
Jebel Ali Free Zone Area, Near Container Terminal 3 - North Zone
Dubai
UAE
00971 4 8014444 Customer Services, 00971 48146800 00971 4
8014000 Reception phone numbers

Nalco Egypt Trading
5th Settlement, South 90th St.
The Address Building No 67th – 1st floor
New Cairo, Cairo, Egypt11835
0020 2 25 37 1195,

Ecolab Maroc S.A.R.L.
Batiment, Lot. Mounir 1, Parc Industriel
Attawfik
Route 1029
Sidi Maarouf, Casablanca, Morocco
00212 22 58 25 30 - 35,

Emergency telephone : +32-(0)3-575-5555
number

Issuing date : 02.04.2019

Section: 2. HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Skin irritation : Category 2
Eye irritation : Category 2B

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS Label element
Product AS SOLD

Hazard pictograms :



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SOLITAIRE

Signal Word : Warning

Hazard Statements : Causes skin and eye irritation.

Precautionary Statements : **Prevention:**
Wash skin thoroughly after handling. Wear protective gloves.
Response:
IF ON SKIN: Wash with plenty of soap and water. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If skin irritation occurs: Get medical advice/attention. If eye irritation persists: Get medical advice/attention. Take off contaminated clothing and wash before reuse.

Product AT USE DILUTION

Precautionary Statements : **Prevention:**
Wash hands thoroughly after handling.
Response:
Get medical advice/ attention if you feel unwell.
Storage:
Store in accordance with local regulations.

Product AS SOLD

Other hazards : None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Organic sulfonic acid salt	25155-30-0	10 - 30
Polyethylene Glycol	25322-68-3	10 - 30
dodecanamide, n-(2-hydroxyethyl)-	142-78-9	10 - 30
Sodium poly(oxyethylene) dodecyl ether sulfate	68585-34-2	5 - 10
Amides, coco, N-[3-(dimethylamino)propyl], N-oxide	68155-09-9	5 - 10
acetic acid, sodium salt	127-09-3	1 - 5
ethanol	64-17-5	1 - 5
ethylene oxide	75-21-8	0.1 - 1
1,4-dioxane	123-91-1	0.1 - 1

Product AT USE DILUTION

No hazardous ingredients

Section: 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Get medical attention if irritation develops and persists.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

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Protection of first-aiders	: No special precautions are necessary for first aid responders.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

Section: 5. FIREFIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during firefighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx) Sulphur oxides Oxides of phosphorus
Special protective equipment for firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Sweep up and shovel into suitable containers for disposal.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Refer to protective measures listed in sections 7 and 8.
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Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. Do not flush into surface water or sanitary sewer system.

Section: 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Avoid contact with skin and eyes. Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
ethanol	64-17-5	TWA OEL-RL	1,000 ppm 1,900 mg/m ³	ZA OEL
ethanol	64-17-5	TWA	1,000 ppm 1,880 mg/m ³	ARE OEL
ethylene oxide	75-21-8	TWA OEL-CL	5 ppm 10 mg/m ³	ZA OEL
		TWA OEL-RL	5 ppm 10 mg/m ³	ZA OEL
ethylene oxide	75-21-8	TWA	1 ppm 1.8 mg/m ³	2004/37/EC
ethylene oxide	75-21-8	TWA	1 ppm 1.8 mg/m ³	ARE OEL
1,4-dioxane	123-91-1	TWA OEL-RL	25 ppm 90 mg/m ³	ZA OEL
		STEL OEL-RL	100 ppm 360 mg/m ³	ZA OEL
1,4-dioxane	123-91-1	TWA	25 ppm 90 mg/m ³	BH OEL
		STEL	40 ppm 135 mg/m ³	BH OEL
1,4-dioxane	123-91-1	TWA (8 Hour)	100 ppm 360 mg/m ³	TR OEL
		TWA (8 Hour)	20 ppm 73 mg/m ³	TR OEL

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1,4-dioxane	123-91-1	TWA	20 ppm 72 mg/m3	ARE OEL
		STEL	90 ppm 135 mg/m3	ARE OEL

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : Due to the form and packaging of the product, no protective equipment is needed under normal use conditions.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: Cast Solid	liquid
Colour	: opaque, light pink	colourless
Odour	: citrus	citrus
pH	: 7.0 - 9.0, (1 %)	7.0 - 8.0
Flash point	: Not applicable.	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: no data available	
Evaporation rate	: no data available	
Flammability (solid, gas)	: no data available	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	

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Vapour pressure	: no data available
Relative vapour density	: no data available
Relative density	: no data available
Water solubility	: insoluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: no data available
Molecular weight	: no data available
VOC	: no data available

Section: 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: None known.
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides nitrogen oxides (NOx) Sulphur oxides Oxides of phosphorus

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes eye irritation.
Skin	: Causes skin irritation.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

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Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Irritation
Skin contact	: Redness, Irritation
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity	: Acute toxicity estimate : 3,188 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : 112.22 mg/l Test atmosphere: dust/mist
Acute dermal toxicity	: Acute toxicity estimate : > 5,000 mg/kg
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: Mild eye irritation
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available
Germ cell mutagenicity	: no data available
Teratogenicity	: no data available
STOT - single exposure	: no data available
STOT - repeated exposure	: no data available
Aspiration toxicity	: no data available

Section: 12. ECOLOGICAL INFORMATION

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Product AS SOLD

Ecotoxicity

Environmental Effects : Toxic to aquatic life.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Organic sulfonic acid salt
96 h LC50 Fish: 3.2 mg/l

Polyethylene Glycol
96 h LC50 Fish: > 1,000 mg/l

dodecanamide, n-(2-hydroxyethyl)-
96 h LC50 Fish: > 100 mg/l

Sodium poly(oxyethylene) dodecyl ether sulfate
96 h LC50 Fish: 28 mg/l

acetic acid, sodium salt
96 h LC50 Fish: > 100 mg/l

ethanol
96 h LC50 Pimephales promelas (fathead minnow): > 100 mg/l

ethylene oxide
96 h LC50: 84 mg/l

1,4-dioxane
96 h LC50 Fish: 1,000 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : ethylene oxide
48 h LC50: 212 mg/l

Components

Toxicity to algae : ethylene oxide
96 h EC50: 240 mg/l

Persistence and degradability

Product AS SOLD

no data available

Product AT USE DILUTION

no data available

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

no data available

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Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

- Disposal methods : Do not contaminate ponds, waterways or ditches with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

- Disposal methods : Diluted product can be flushed to sanitary sewer.
- Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR/ADN/RID)

- UN number : Not dangerous goods
- UN proper shipping name : Not dangerous goods
- Transport hazard class(es) : Not dangerous goods
- Packing group : Not dangerous goods
- Environmental hazards : Not dangerous goods
- Special precautions for user : Not dangerous goods

Air transport (IATA)

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

Section: 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

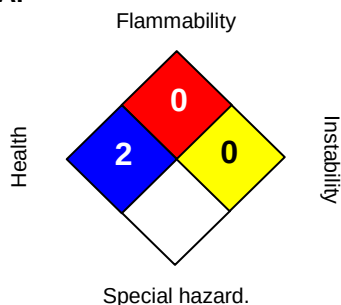
SAFETY DATA SHEET

SOLITAIRE

Section: 16. OTHER INFORMATION

Product AS SOLD

NFPA:



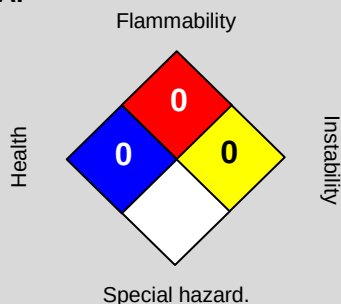
HMIS III:

HEALTH	2
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 02.04.2019
Version : 1.2
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.